

MANTEY JOSHUA KOJO ADUAMPONG

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TECHNICAL SKILLS

MD & Simulation: AMBER, NAMD, GROMACS, Schrodinger, VMD, PyMOL, ChimeraX, MDAnalysis, MDTraj

Programming & Data: Python, MATLAB, Bash, MongoDB, SQLite, FastAPI, Plotly Dash, LangChain, FAISS, OpenAI SDK, Biopython, Streamlit, NCBI E-utilities, NSF ACCESS, SLURM

RESEARCH AND COMPUTATIONAL EXPERIENCE

Graduate Research Assistant, Computational Biophysics

Jan 2025 – Present

Northeastern University | Boston, MA

- Designed and executed large-scale MD simulations of MUC2 mucin (5,200+ amino acids, 3M+ atom systems) using AMBER and GROMACS on NSF ACCESS HPC clusters, generating 10+ TB of trajectory data across 1.2 μ s simulations.
- Developed end-to-end Python pipelines implementing cross-normalized DCCM, PCA, t-SNE, and spectral embedding to compare glycosylated vs. non-glycosylated MUC2 systems with direct implications for drug permeation.

CIML Summer Institute, Deep Learning and Neural Networks

Summer 2026

UC San Diego / Computational Immunology and ML Institute | San Diego, CA

- Intensive training in deep learning architectures (CNNs, RNNs, transformers) applied to high-dimensional biological datasets; implemented neural network workflows for sequence modeling and pattern recognition.

Research Collaborator, HIV Vaccine Development

May – July 2025

Fred Hutchinson Cancer Center | Remote

- Analyzed 8+ conformational states from MD simulations and cryo-EM structures of HIV envelope proteins (gp120, gp41) to identify conserved binding pockets for broadly neutralizing antibody development.

Computational Modeling and Bioprocess Projects

Sept 2024 – May 2025

Northeastern University | Boston, MA

- Built ML models on 50,000+ atomic configurations to predict CO₂ adsorption on NH₄HCO₃ for industrial carbon capture; processed CFD data (1M+ mesh elements) for natural gas storage optimization.
- Achieved 250 mg/L recombinant protein yield applying statistical process control and DOE to bioreactor data; reduced batch variability by 30% via PAT framework with real-time sensor monitoring.

DATA SCIENCE AND ML DEPLOYMENT PROJECTS

Diabetes Risk Predictor | XGBoost, SHAP, Streamlit

2025 – 2026

- Built an end-to-end ML pipeline on the Pima Indians dataset (768 patients); benchmarked 10 classification models using cross-validation and F1-optimized thresholding, achieving ROC-AUC of 0.85+.
- Integrated SHAP TreeExplainer for per-patient feature attribution; deployed interactive Streamlit web app with real-time risk prediction, visual gauge, and clinical feature summaries.

Biomedical Literature RAG Pipeline | PubMed, FAISS, LangChain, OpenAI

2025 – 2026

- Built an end-to-end RAG pipeline over approximately 1,000 PubMed clinical trial abstracts using OpenAI text-embedding-3-small embeddings and a FAISS IndexFlatIP vector index for biomedical literature retrieval.
- Engineered MeSH-aware XML ingestion from NCBI Entrez with PMID-grounded citations; designed a strict grounding prompt enforcing inline references for trustworthy deployment in regulated biomedical domains.

SELECTED PUBLICATIONS AND PRESENTATIONS

- Amevor, R., Mantey J.K.A., Nketiah, T.A. "Evaluation of AI-Based Predictive Models for Early Cancer Detection." *Jr.Inn.Sci* 2.1 (2026).
- Mantey J.K.A., Muntaka I. "Economic Evaluations of Advanced Drug-Delivery Platforms." SARC Publisher, 2025.
- Mantey J.K.A., Meteku B.E., Baffoe E. "Amino Acids as Corrosion Inhibitors." Taylor & Francis Group, 2025.
- Mantey J.K.A. et al. "Enhanced Sampling to Study MUC2 Interactions with Bile Salts." AIChE Annual Meeting, Boston. Nov 2024. [Oral]

AWARDS, LEADERSHIP AND ENTREPRENEURSHIP

- People's Choice Award for Outstanding Impact and Community Recognition, **World Bank Group Youth Summit 2026**, Washington DC (1 of 6 finalists from 700+ teams across 190 countries).
- Co-Founder and Group Leader, Sankofa Fresh, AI-powered digital marketplace connecting Ghanaian smallholder farmers to buyers; **World Bank Youth Summit** finalist 2026.
- Academic and Networking Chair and Mentorship Chair, African Graduate Student Association (AGSA), Northeastern University.

EDUCATION

Ph.D., Chemical Engineering, *Northeastern University*

Sept 2024 – Present

M.S., Biosystems and Biomaterials, *Northeastern University*

Sept 2024 – May 2026

Applied Data Science Certificate, *WorldQuant University*

2026

B.S., Petrochemical Engineering, *Kwame Nkrumah University of Science and Technology (KNUST)*

Sept 2019 – Sept 2023